



# Escaping the CS Dungeon: Modern College Curricula within and Beyond Computing

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# Knowledge Gap

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- Need for computing skills
- Lack of documentation
- CS courses for non-majors
  - How common are they?
  - What fields are supported?
  - Which programming language?
  - Which departments/institutions offer them?

# Introduction

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- How/where computing is integrated
- Computing coursework
  - Which disciplines
  - Use cases
  - Techniques
- Support future research

# Background

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**Growing demand  
for knowledge of  
basic STEM  
concepts**

(Chilana et al., 2015)

**Rise of interest in  
programming  
among non-CS  
students**

(Ruvolo et al., 2019)

**Fostering non-  
STEM interest in  
STEM is still  
needed**

(Shin et al., 2016)

# Methodology – Course Selection

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- Top 50 largest public universities (NCES)
- **Only** 4-year large public universities
  - No community colleges
  - No small/private institutions
- Courses
  - Keywords used: Java, C++, C#, C, Python, JavaScript, and Programming
  - Within previous year
  - No prerequisites

# Methodology – Survey

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- Online Survey
- Instructor self-reported data:
  - Course name/department
  - Average number of students enrolled
- Course structure questions:
  - Programming language(s) used in the course
  - Environments/tools used for instruction
  - Strengths/weaknesses of tools/environments
  - Instructor policies on collaboration
- 132 contacted, 81 responses

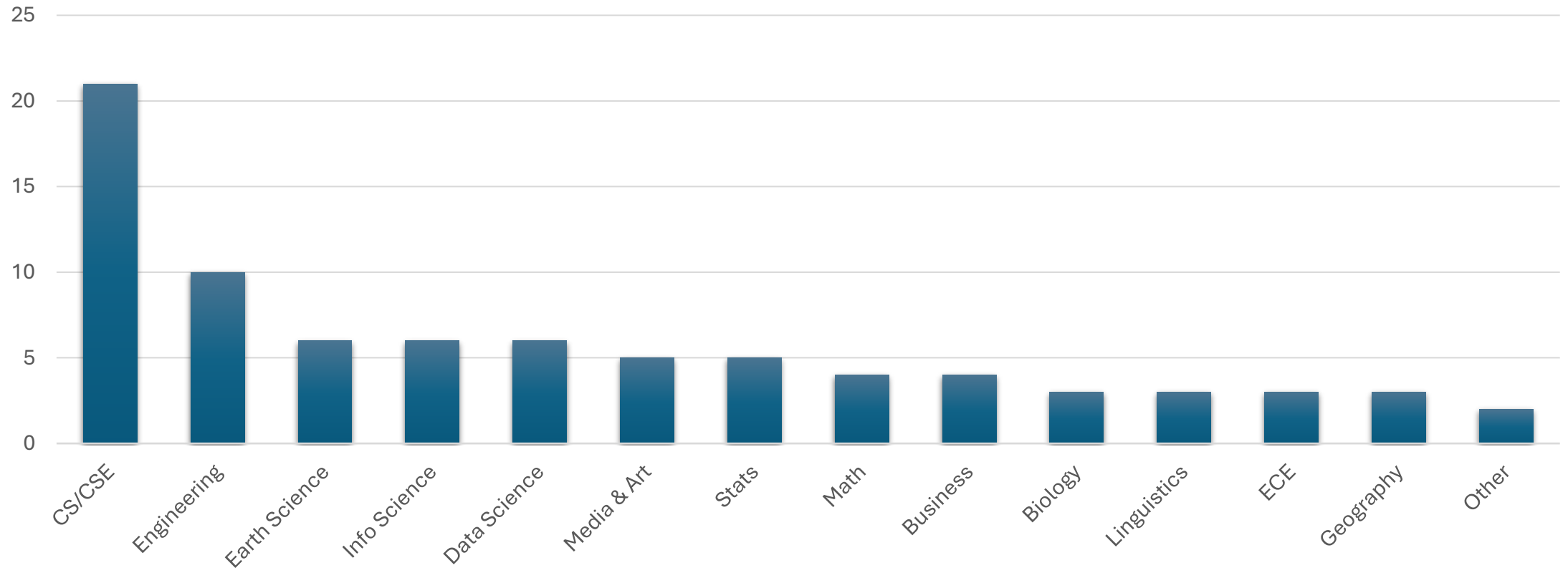
# Findings – Course Categories

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- Disciplines
  - Based on department
  - Similar departments grouped
  - Outliers grouped by degrees
- Categories
  - Computing, STEM, non-STEM
  - Disciplines represented in categories
- Computing course variety

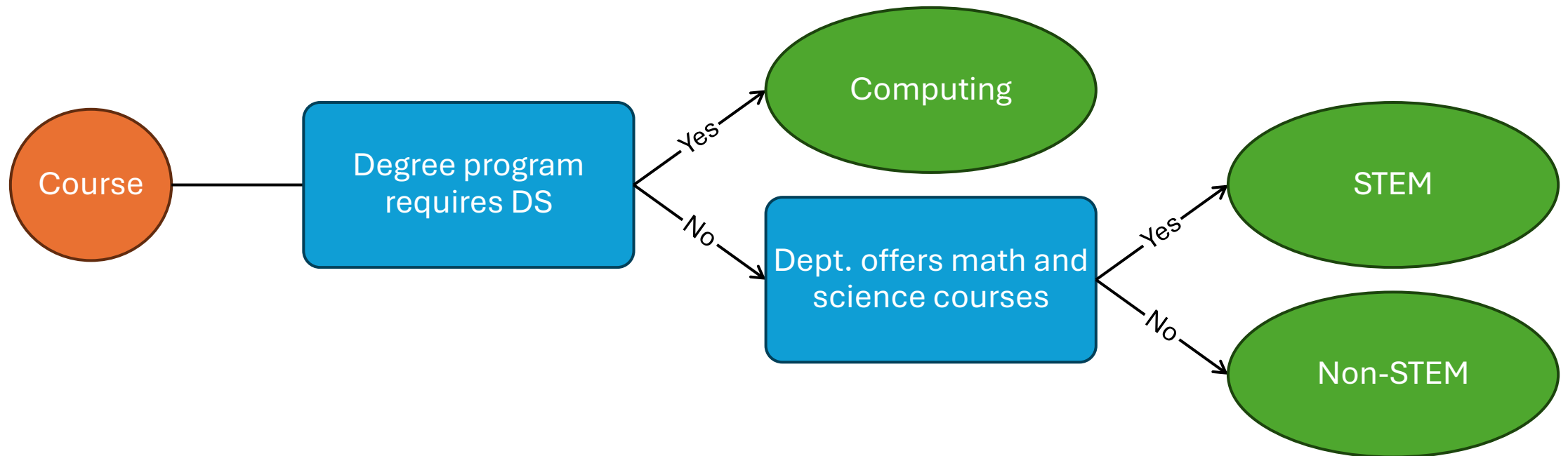
# Course Disciplines

Disciplines Represented in all Courses (N=81)



# Course Categorization Methodology

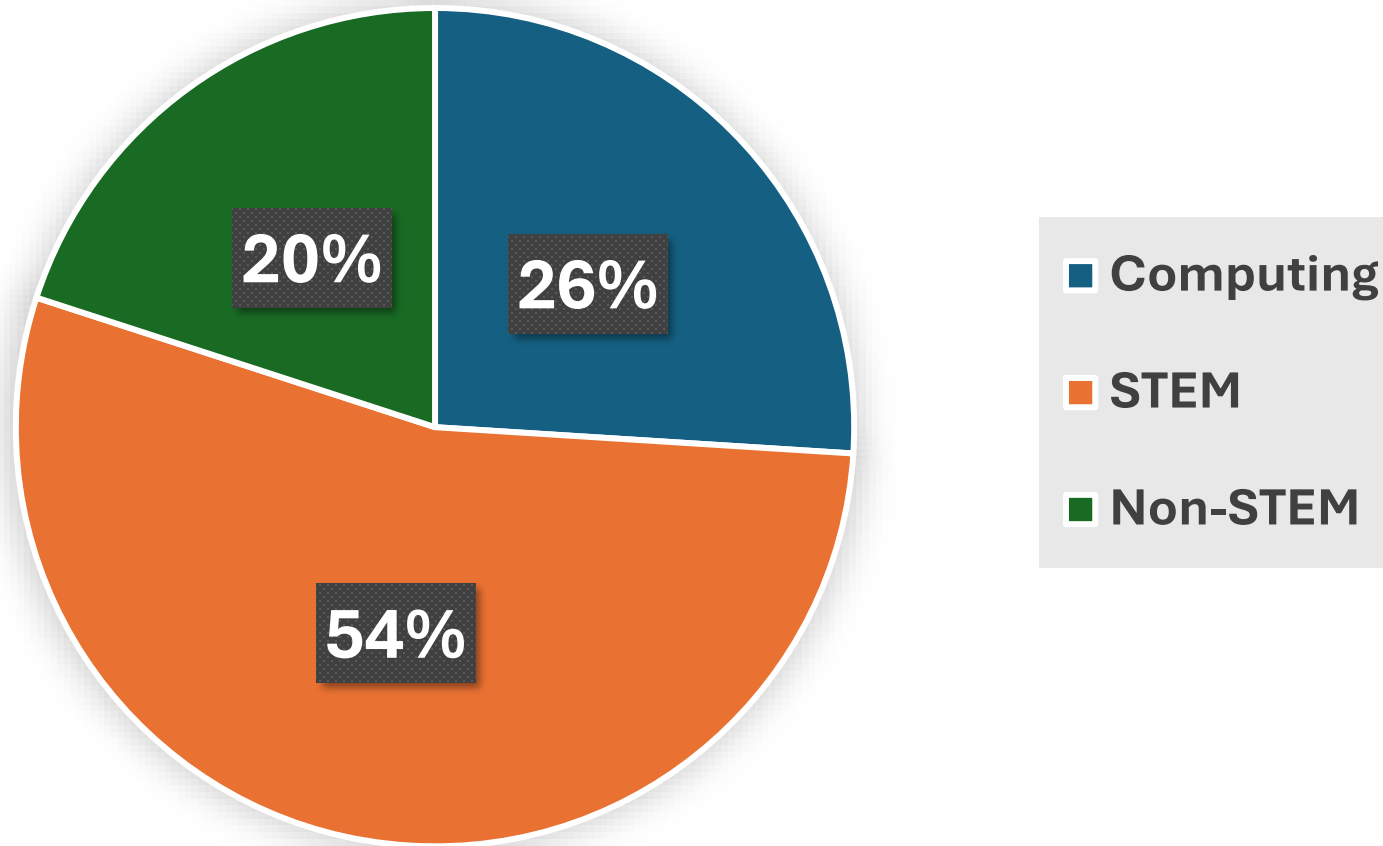
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# Course Categorization

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## Course Categories

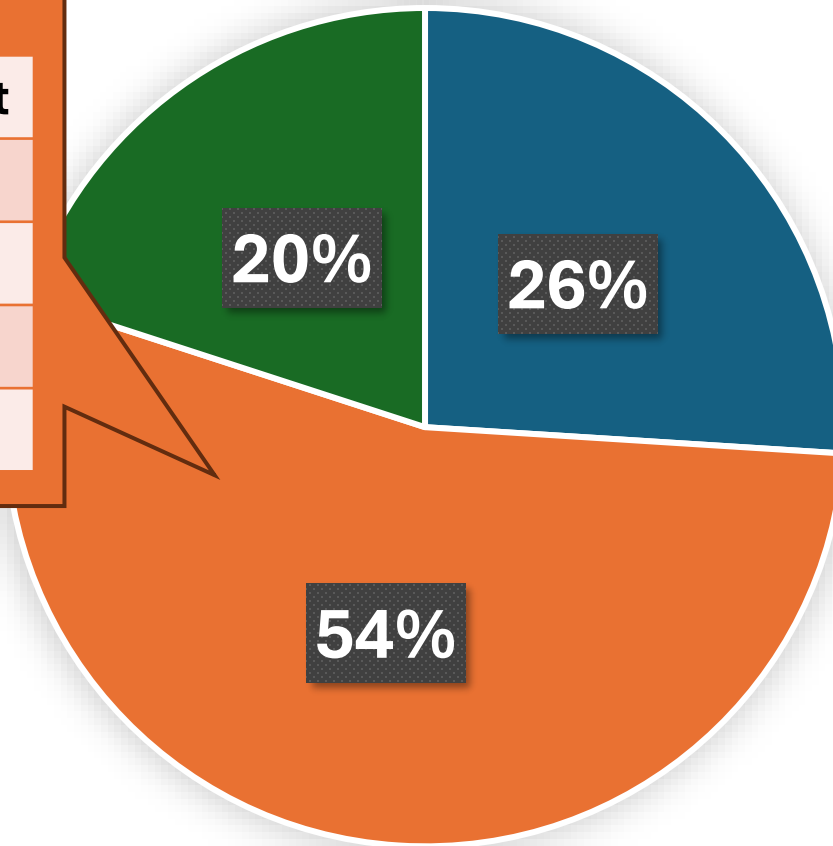


# Course Categorization

## Disciplines Represented in STEM (N=44)

| Discipline    | Count |
|---------------|-------|
| Engineering   | 10    |
| Earth Science | 6     |
| CS/CSE        | 5     |
| ...           |       |

## Course Categories



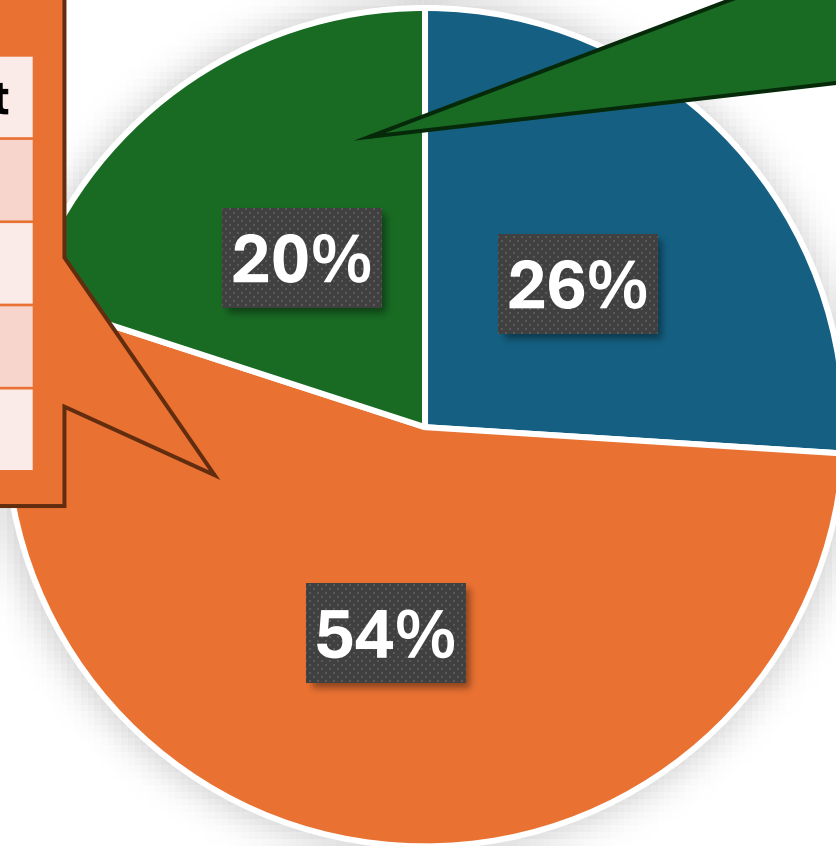
- Computing
- STEM
- Non-STEM

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## Course Categories



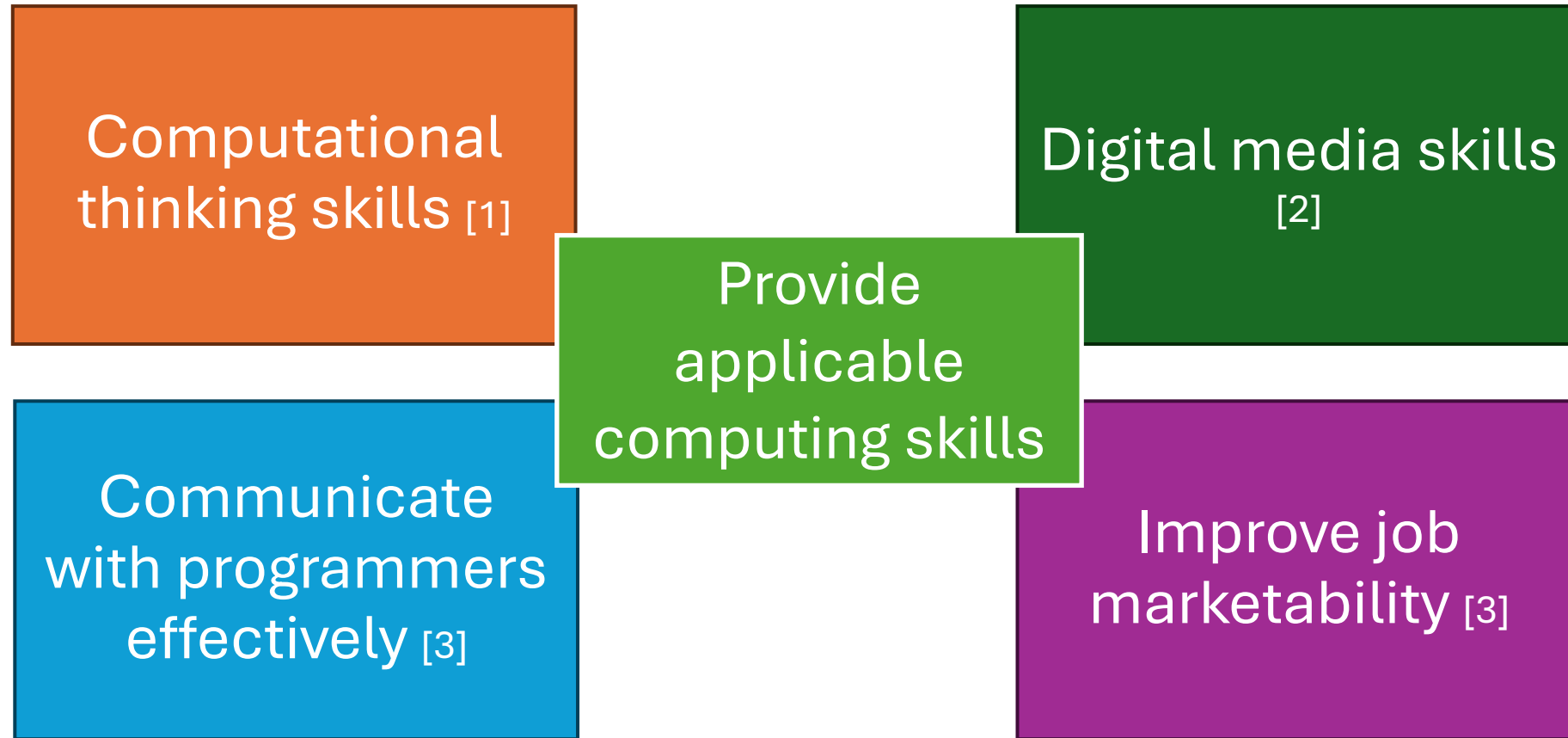
## Disciplines Represented in non-STEM (N=16)

| Discipline  | Count |
|-------------|-------|
| Media & Art | 5     |
| Business    | 3     |
| Linguistics | 3     |
| ...         |       |

- Computing
- STEM
- Non-STEM

# Programming for Non-Majors

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1. Prinsley & Johnston, 2015

2. Peppler & Kafai, 2009

3. Chilana et al., 2015

# Findings – Course Structure

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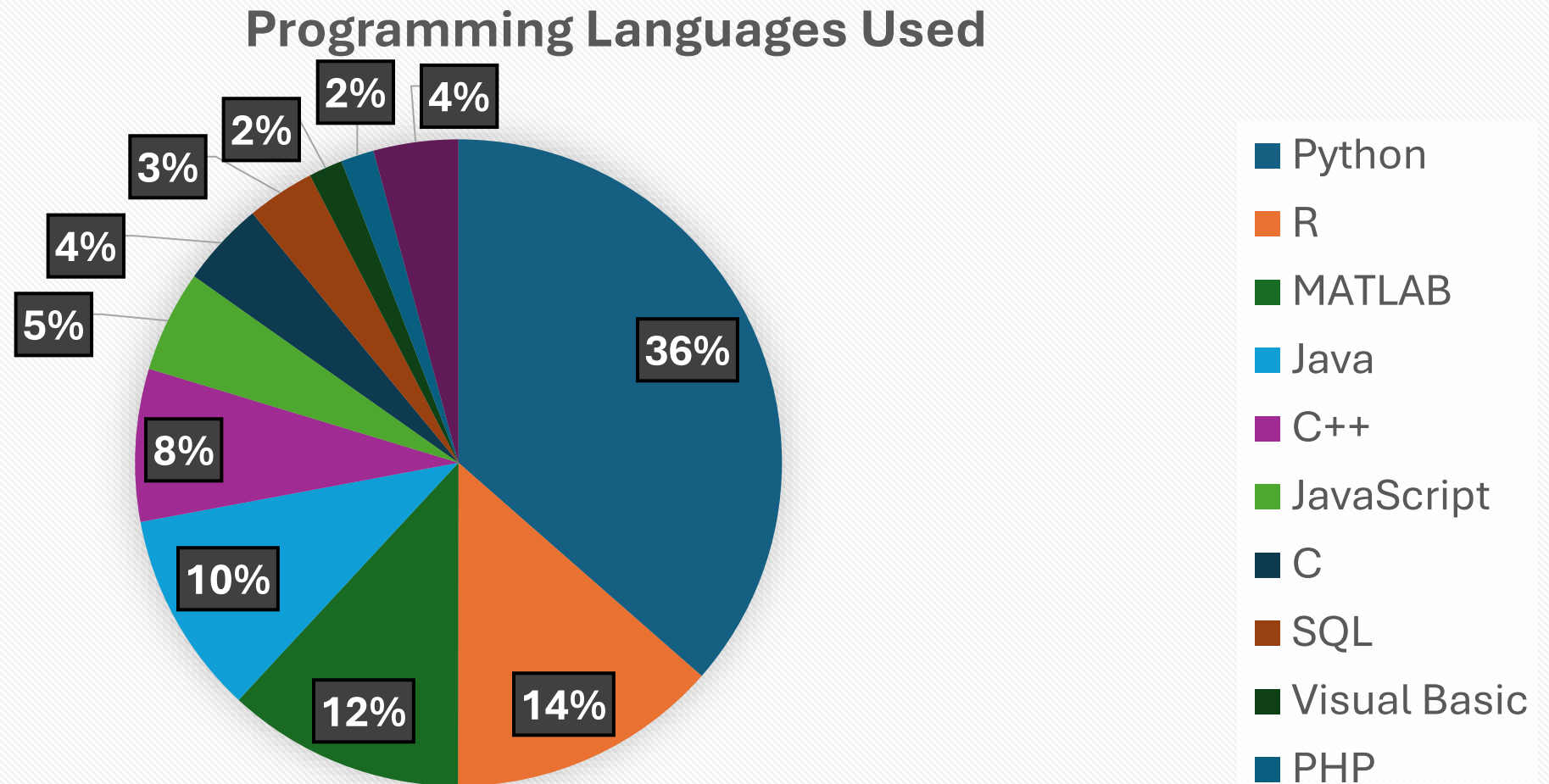
- Trends in course structure
  - Programming languages used
  - Course collaboration policies
- Differences between categories?

# Tailored Programming Courses

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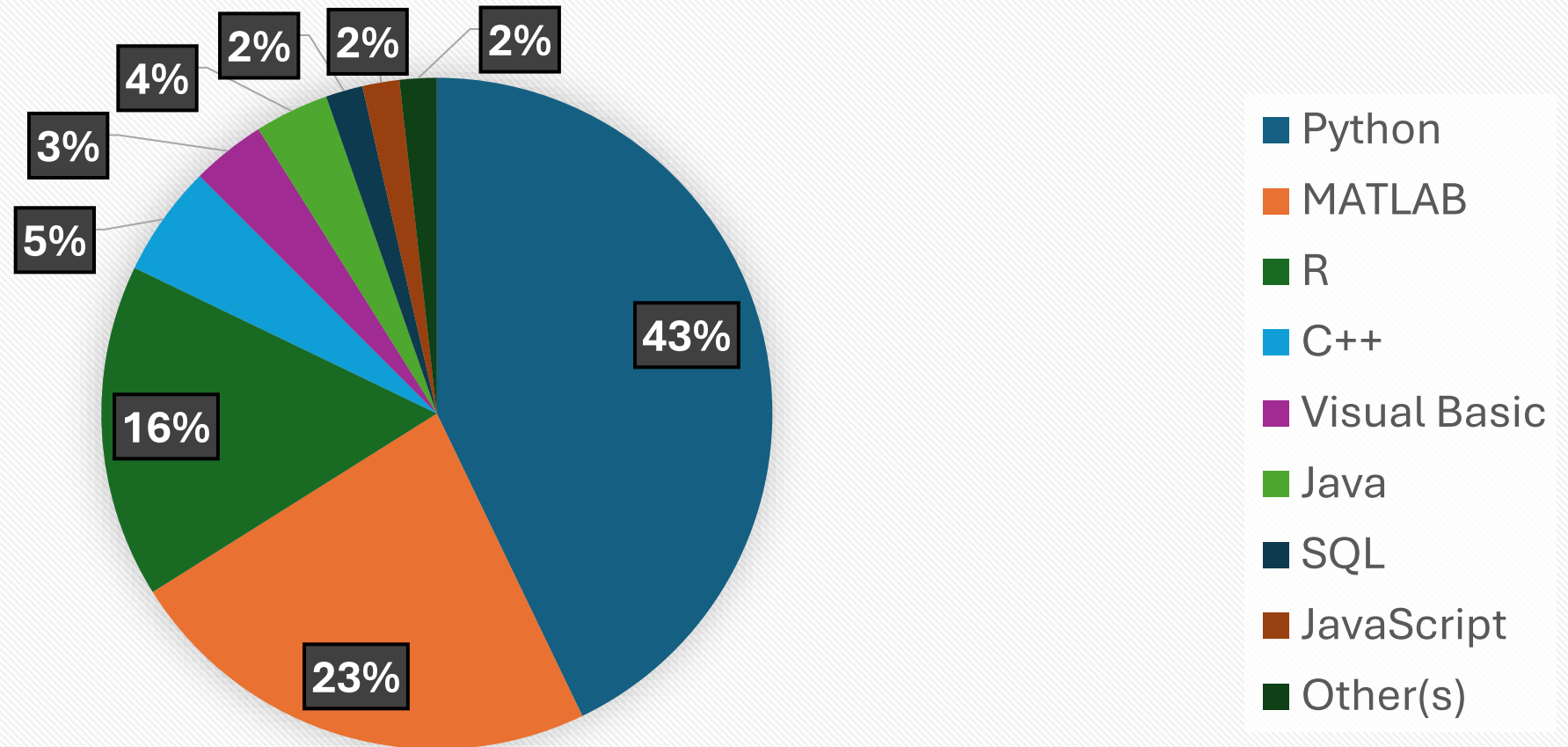
- Discipline tailored courses
  - STEM Ex: “Programming for Biologists”
  - Non-STEM Ex: “Computational Techniques for Finance”
- Higher motivation and engagement
  - Non-STEM Ex: media computation and art software

# Programming Languages



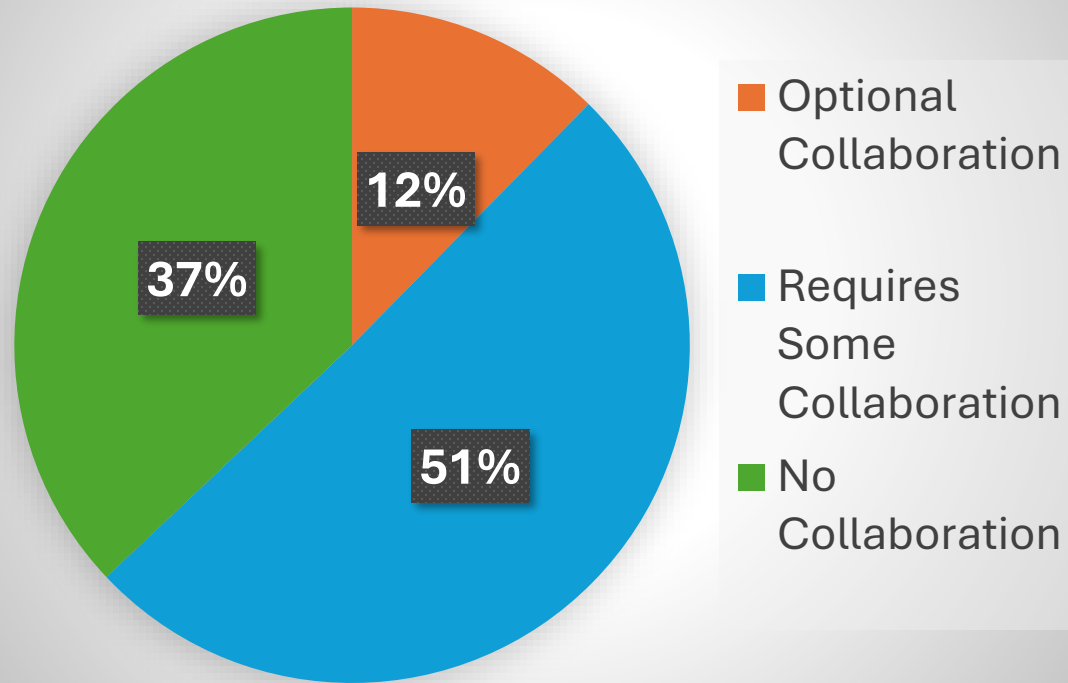
# Programming Languages – STEM

Programming Languages Used in STEM Courses

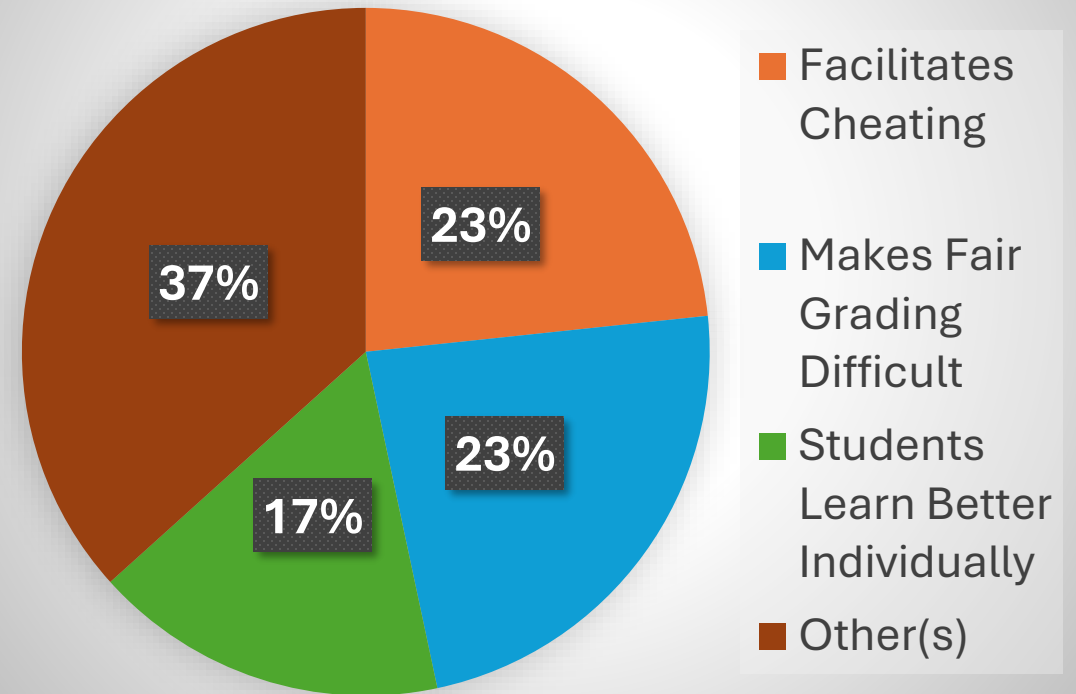


# Instructor Policies on Collaboration

## Level of Collaboration



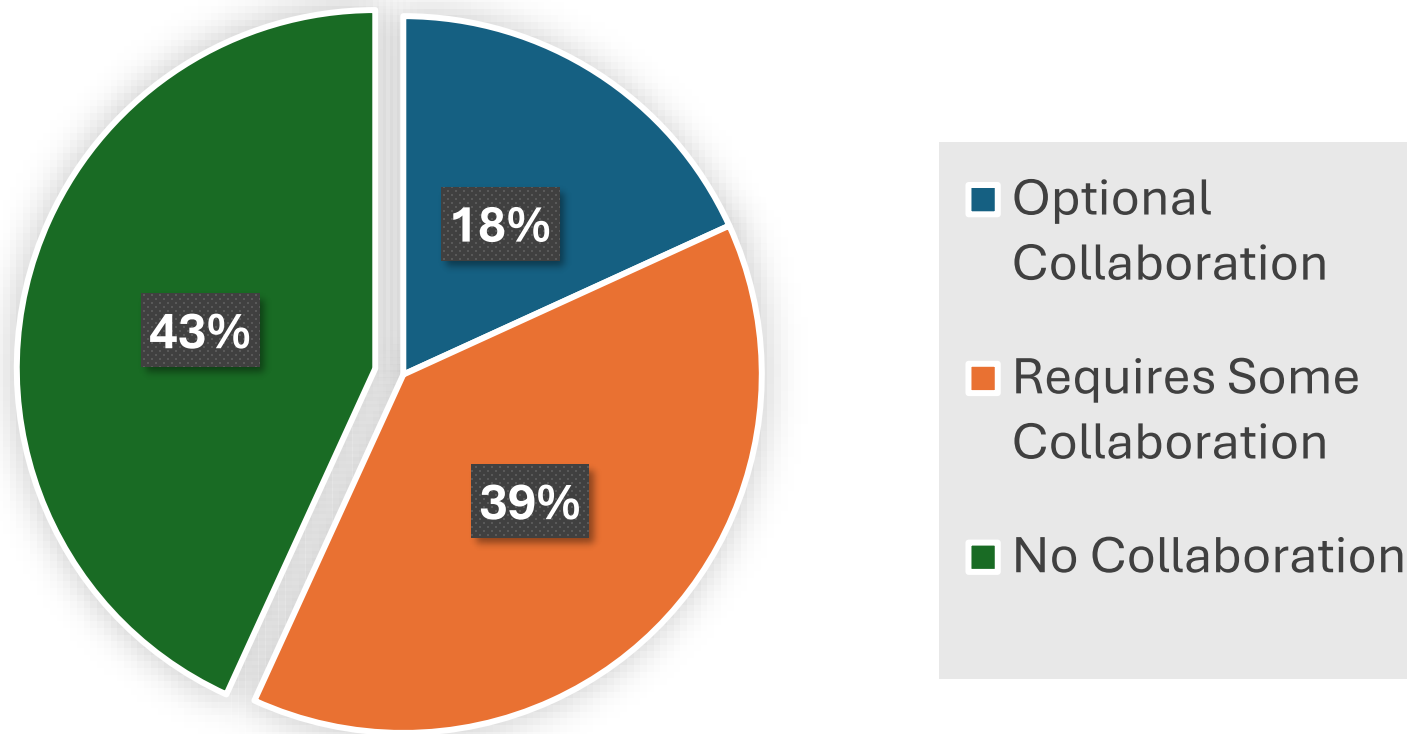
## Reasons for not Allowing Collaboration



# STEM Instructor Policies on Collaboration

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## Percentages of Level of Collaboration in STEM Courses



# Limitations

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- Only represents large 4-year public universities
- Lacks understanding of instructor policies
- Lacks student feedback
  - What formats work best?

# Future Work

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- Applying observations
  - STEM and non-STEM courses
  - Analyze perspectives on course performance
- Follow-up study
  - Collaborative coursework

# References

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# Contact

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